



EEG and clinical predictors of medically intractable childhood epilepsy

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Abstract

Objectives: To identify electroencephalographic and clinical factors associated with both seizure control and medical intractability in children with epilepsy.

Methods: We retrospectively reviewed EEGs and medical records from children with well-controlled epilepsy or medically intractable epilepsy.

Subjects: Features of the initial EEG and clinical findings were compared in 39 children with well controlled seizures and 144 with intractable epilepsy using both univariate and multivariate analyses.

Results: Strong univariate associates were noted between intractability and several EEG factors: abnormal EEG background including diffuse slowing, asymmetry, abnormal amplitude, a high frequency of spikes or sharp waves, and focal spike and wave activity. With multiple logistic regression, independent predictors of intractability were diffuse slowing and focal spike and wave activity. Strong univariate associates of clinical factors with intractability included: an early age of onset, simple partial, tonic, and myoclonic seizures, a history of status epilepticus, a symptomatic etiology of the seizures, and abnormal magnetic resonance imaging of the head. Multivariate analysis detected 4 independent clinical features associated with intractable epilepsy: symptomatic etiology, tonic seizures, simple partial seizures, and an early age of onset.

Conclusions: There are a number of EEG and clinical features that can be identified early in the course of childhood epilepsy that are predictive of outcome. These findings will need to be verified in a prospective study. © 1999 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Intractability; Antiepileptic drugs; Developmental delay; Seizures

1. Introduction

Epilepsy is one of the most common, yet frightening neurological conditions to occur in children. In the United States, there are approximately 125 000 new cases of epilepsy each year; 30% of this group will be less than 18 years old at the time of diagnosis (Hauser and Hersdorffer, 1990; Hauser, 1995). Fortunately, most children with epilepsy do well when treated with an antiepileptic drug (AED) (Camfield et al., 1993; Berg et al., 1995; Carpay et al., 1998). However, a significant number of children with epilepsy do not respond well to AEDs and continue to have seizures (Huttenlocher and Hapke, 1990). While this is a small group of children, the medical, social, and economic consequences of poorly controlled seizures can be enormous. In addition to the unpredictable occurrence of

seizures, these children are often subject to drug toxicity from polytherapy and they are at high risk for behavioral and academic difficulties (Farwell et al., 1985; Holmes, 1997). Uncontrolled seizures may be associated with slowing or a decline in mental development over time (Bourgeois, 1998).

Identifying children destined to develop intractable epilepsy early in their course may be useful in designing therapeutic options. For example, early surgical intervention may be appropriate for children likely to be intractable to antiepileptic drugs. In this study we attempted to identify clinical and EEG factors that predicted medical intractability in newly diagnosed children with epilepsy.

2. Methods

2.1. Medical record review – clinical features

Medical records were received from approximately 1000 children (<18 years) currently followed in the Division of

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Epilepsy and Clinical Neurophysiology at Children's Hospital in Boston. Information was abstracted retrospectively from the child's medical records. To obtain missing information, the child's neurologist was contacted.

Epilepsy was defined as two or more non-provoked seizures. Medical intractability was defined as continued seizures despite adequate trials of 3 or more antiepileptic drugs (AEDs), used either alone or in combination therapy. A trial of AEDs was considered adequate if, in the opinion of the child's neurologist, the dosage was increased until there was either seizure control or signs of clinical or laboratory toxicity. A drug used to treat status epilepticus in the emergency setting was not counted as one of the 3 AEDs. A drug was not considered a failure if serum levels were subtherapeutic or there was documentation of poor compliance. Well-controlled epilepsy was defined as having no seizures for a minimum of one year during the follow-up period.

We included children who had been followed regularly for at least two years. Cases with shorter periods of follow-up or in which there was inadequate follow-up were excluded, with the exception of 8 patients who underwent surgery for catastrophic epilepsy within the two year period (3 with Sturge-Weber Syndrome; 3 with Rasmussen's encephalitis; one with Aicardi syndrome; and one with hemimegalencephaly). While we cannot be absolutely certain that the epilepsy would not have eventually resolved, we felt that the clinical and EEG information from this group of children with particularly severe epilepsy was warranted. We also excluded children who did not have EEGs as part of their initial evaluation. While the majority of children were initially evaluated at Children's Hospital, we included some cases who had their first EEG elsewhere if the clinical information and laboratory studies were reliable and adequate. Although many of the children had multiple EEGs, we only used information from the first EEG obtained after the diagnosis of epilepsy was established.

2.2. Medical record review - EEG features

Electroencephalogram recordings were obtained with a 21 channel electroencephalograph, employing scalp electrodes placed according to the international 10-20 system. In the majority of children, sleep and photic stimulation was obtained. When the child was cooperative hyperventilation was also obtained.

Diffuse slowing was defined as slower than expected background frequencies for the patient's age (Blume, 1982). Excessive beta was defined as greater than expected amounts of beta during the awake state. Asymmetrical background was defined as a 50% or greater difference in amplitude between the two hemispheres. Asynchronous background was defined as a 50% or greater asynchrony in sleep spindles or posterior dominant rhythm. Amplitude was coded as high or low based on published norms for the patient's age (Blume, 1982). Frequency of spikes or sharp waves was coded as greater or less than 1 per 60 s.

Table 1
Factors examined

Electroencephalographic factors	Clinical factors
EEG background	Age of seizure onset
Focal slowing	Age of first EEG
Diffuse slowing	Follow-up duration
Burst suppression	Sex
Excessive beta	Seizures types
Background symmetry	Specific epileptic syndrome
Background synchrony	Seizure frequency
Background amplitude	Status epilepticus
Frequency of sharp wave/spike	Febrile seizures before 1 st unprovoked seizure
Multifocal sharp wave/spike	Family history of seizures
Frequency of sharp wave/spike	Etiology (symptomatic/cryptogenic)
Location of spike and wave	Brain MRI
Reactivity of epileptiform activity to sleep	Antiepileptic drugs used (number)
Reactivity of epileptiform activity to hyperventilation	
Reactivity of epileptiform activity to photic stimulation	

Multifocal EEGs were those with two or more independent epileptic foci.

We collected 144 cases of intractable cases and 39 cases of well-controlled cases from the above review of charts. The excluded cases were those patients in which seizure control did not meet the definition of either intractable or well-controlled cases or in whom there was inadequate follow-up.

Table 1 lists the electroencephalographic and clinical features at the onset of the epilepsy which were examined in this study. The seizure frequency listed refers to the number of seizures the child had during the initial phase of the disorder. By definition, the well-controlled group had at least one year of seizure freedom during the follow-up period.

2.3. Data analysis

Data analysis was obtained using SAS software. Data were analyzed with chi-square and the chi-square test for trend for univariate comparisons. Fisher's exact tests were used for univariate comparisons when there were no cases in a category. Odds ratios, 95% confidence intervals, and *P*-values were computed from univariate and multivariate logistic regression models.

In the univariate analysis we performed hypothesis testing for each of the potential predictors. This process involves a possibility of observing statistical significance in some of the predictors purely by chance, because of multiple testing. One approach to this problem may be applying the Bonferroni correction. However, we present the original *P* values and use a significance level of 0.05 without correction, because univariate analyses reflect an

Table 2
Characteristics of intractable and well-controlled group

	Intractable (<i>n</i> = 144)	Controlled (<i>n</i> = 39)
Age of seizure onset (mean, year)	2.9	5.2
Age of first EEG (mean, year)	5.8	6.8
F/U duration (mean, year)	5.3	4.9
Male sex (%)	73 (50.7)	21 (53.9)
Number of antiepileptic drugs (mean)	8.1	2.7

exploratory analysis. We considered these *P* values as suggestive, rather than conclusive, evidence for association, as well as a means of comparing relative importance among the predictors in order to select candidates for multivariate analysis. Following univariate analysis we performed multivariate analysis to choose independently significant factors among the many significant factors in univariate analysis. We discarded some factors in performing the multivariate analysis because of a zero cell.

Among potential clinical predictors, we discarded the Lennox–Gastaut syndrome because of the zero cell; symptomatic etiology since it overlapped with an abnormal brain MRI, and seizure frequency since it was influenced by age of seizure onset.

3. Results

3.1. Patient profiles

The clinical profiles of the patients with intractable and well-controlled group are presented in Table 2. There were no differences of age at the time of the first EEG, duration of follow-up, and sex between the two groups. However, the age of seizure onset in the intractable group (mean = 2.9 year) was younger than that in the well-controlled group (mean = 5.2 year), and the risk of intractability decreased with age (OR = 0.83 per year, *P* < 0.001). Duration of follow-up and sex distribution was similar in the two groups. The mean numbers of AEDs tried were 8.1 in the intractable group and 2.7 in the well-controlled group, and the difference was significant (*P* < 0.001).

3.2. EEG features

Comparison of the two groups demonstrated strong univariate associations between intractability and several initial EEG features (Table 3). The frequency of abnormal EEG background was higher in the intractable group; diffuse slowing was the most significant factor (*P* = 0.01). Asymmetries were noted only in the intractable group (*P* = 0.02) and asynchronous background was observed more frequent in the intractable group (*P* = 0.11). High or

Table 3
Description of Electroencephalographic features^a

	Intractable (<i>n</i> = 144)	Controlled (<i>n</i> = 39)	OR	95% CI	<i>P</i> -value
Abnormal EEG background (%)	106 (73.6)	16 (41.0)	4.01	1.93, 8.52	< 0.001
Focal slowing	44 (30.6)	7 (18.0)	2.01	0.87, 5.27	0.12
Diffuse slowing	71 (49.3)	10 (25.6)	2.82	1.32, 6.48	0.01
Burst suppression	8 (5.6)	1 (2.6)	2.24	0.39, 42.12	0.45
Excessive beta	9 (6.3)	0 (0.0)			0.21*
Asymmetric background (%)	19 (13.2)	0 (0.0)			0.02*
Asynchronous background (%)	18 (12.5)	1 (2.6)	5.43	1.07, 99.20	0.11
Abnormal amplitude (%) (high or low)	34 (23.6)	3 (7.7)	3.71	1.24, 16.04	0.04
Frequency of sharp wave/spike (%) > 1/60 s	75 (52.1)	9 (23.1)	3.62	1.66, 8.60	< 0.01
Multifocality of sharp wave/spike (%)	66 (45.8)	12 (30.8)	1.90	0.91, 4.17	0.09
Frequency of spike and wave (%)					
< 2.5 Hz	21 (14.6)	1 (2.6)	5.34	1.05, 97.55	0.11
3 Hz	5 (3.5)	8 (20.5)	0.16	0.05, 0.51	< 0.01
Location of spike and wave (%)					
Generalized	52 (36.1)	16 (41.0)	0.81	0.40, 1.70	0.57
Frontal predominant	22 (15.3)	2 (5.1)	3.34	0.92, 21.42	0.11
Focal	32 (22.2)	2 (5.1)	5.29	1.50, 33.59	0.03
Reactivity of epileptiform activity					
to sleep (%) ^a	39 (40.0)	8 (22.2)	1.57	0.68, 3.97	0.31
to hyperventilation (%) ^b	7 (9.7)	7 (24.1)	0.34	0.10, 1.09	0.07
to photic stimulation (%) ^c	3 (2.5)	6 (15.8)	0.14	0.03, 0.54	< 0.01

^a OR, odds ratio; CI, confidence interval, *P*-values marked by * are obtained by Fisher's exact test because of zero cell. Missing values (a, b, c) are as follows because provocation tests could not be performed in some patients; (a) 21, (b) 82, and (c) 24 cases.

Table 4
Results of multivariate analysis for predictors of intractable childhood epilepsy

Predictors	Odds ratio	95% CI	P-value
<i>Electroencephalographic factors^a</i>			
Photoconvulsive response	0.10	0.02, 0.51	0.009
3 Hz spike and wave	0.21	0.05, 0.74	0.016
Diffuse slowing	2.59	1.05, 6.84	0.045
Focal spike and wave	5.21	1.25, 38.16	0.049
Frequent sharp wave/spike (> 1/60 s)	1.68	0.67, 4.43	0.277
<i>Clinical factors^b</i>			
Symptomatic etiology	4.41	1.68, 13.00	0.004
Tonic seizure	13.50	2.56, 249.74	0.014
Age of seizure onset	0.86	0.75, 0.97	0.016
Simple partial seizure	5.11	1.21, 36.08	0.049
Status epilepticus	7.70	1.42, 143.48	0.055
Absence seizure	0.92	0.32, 2.70	0.881

^a Factor which was not included in this model but thought to be significant, asymmetric background.

^b Factor which was not included in this model but thought to be significant, Lennox–Gastaut Syndrome.

low amplitude activity was seen more commonly in the intractable group ($P = 0.04$).

The number of sharp waves and spikes were also different in the two groups, with children with intractable epilepsy more likely to have frequent discharges (> 1/60 s) than those children with well-controlled epilepsy. Children in the intractable group were more likely to have multifocal sharp waves and spikes than those with well-controlled seizures, but failed to reach statistical significance. Generalized discharges with a frequency of < 2.5 Hz, while more common in the intractable group, did not reach statistical significance; 3 Hz spike and wave discharges were more commonly present in the well-controlled group ($P < 0.01$). Focal spike and wave discharges were more common in the medically intractable group ($P = 0.03$).

Surprisingly, multifocal EEGs were not associated with medical intractability. However, benign partial epilepsies, such as benign Rolandic epilepsy, with two or more independent foci, was classified into the multifocal group. This may be the reason why multifocal EEGs were not significantly associated with medical intractability.

Among the provocation procedures, photic stimulation was the only statistically significant factor ($P < 0.01$); it was seen more frequently in the well-controlled group. All 6 of the 38 (15.8%) children with photoconvulsive responses and well-controlled seizures had other abnormalities on the EEG; 4 had normal background and interictal spike and wave activity and two had abnormal background with interictal spike and wave activity. All of the 3 patients who had photoconvulsive responses and medical intractability had other abnormalities (normal background and interictal spike and wave in one, abnormal background and

interictal spike and wave in two). A photoconvulsive response therefore was a favorable factor regardless of other EEG abnormalities.

Multivariate analyses were performed to assess the association between intractable epilepsy and potential predictive clinical factors and EEG factors (Table 4). Diffuse slowing (OR = 2.59, 95% CI 1.05, 6.84) and focal spike and wave (OR = 5.21, 95% CI 1.25, 38.16) were independent predictors of intractable epilepsy, while frequent sharp waves and spikes (> 1/60 sec) were no longer significant in multivariable analyses. Asymmetric background could not be included in the multivariable model because of the zero cell. Reactivity to the photic stimulation (OR = 0.10, 95% CI 0.02, 0.51) and 3 Hz spike and wave discharges (OR = 0.21, 95% CI 0.05, 0.74) were independent predictors of well controlled epilepsy.

3.3. Clinical features

A comparison of two groups in regards to clinical factors showed several differences statistically (Table 5). Simple partial, tonic, and myoclonic seizures were more frequent in the intractable group, while absence seizures were more frequent in the well-controlled group. Among the specific epileptic syndrome, Lennox–Gastaut Syndrome was found only in the intractable group, while the frequency of infantile spasm was higher in the intractable group, but failed to reach statistical significance ($P = 0.09$). Seizure frequency was higher in the intractable group. While status epilepticus was much more common in the intractable group, febrile seizures and family history of seizures were not significant. A symptomatic etiology and abnormal brain MRI were more common in the intractable group.

Among the other clinical factors, motor developmental delay, motor deficit, and mental retardation were found more frequently in the intractable group. Language development delay, microcephaly, neonatal seizure, and behavioral problems were reported more commonly in the intractable group but failed to demonstrate statistical significance.

Using multivariate analysis, symptomatic etiology (OR = 4.41, 95% CI 1.68, 13.00), tonic seizures (OR = 13.50, 95% CI 2.56, 249.74), age of seizure onset (OR per year = 0.86, 95% CI 0.75, 0.97), and simple partial seizures (OR = 5.11, 95% CI 1.21, 36.08) were independent predictors of intractable epilepsy, while status epilepticus and absence seizures were no longer significant in multivariable analyses (Table 4). The potential predictors, seizure frequency and brain MRI, were explained by the other factors (age of seizure onset and symptomatic etiology) in the multivariable model. Lennox–Gastaut syndrome could not be included in the multivariable model because of the zero cell.

Table 6 compares the electroclinical syndromes identified in this study in the well-controlled and intractable groups. As can be seen, patients with infantile spasms and Lennox–

Table 5
Description of clinical factors^a

	Intractable (n = 144)	Controlled (n = 39)	OR	95% CI	P value
Age of seizure onset (year)			0.83	0.75, 0.92	< 0.001
Seizure types (%)					
SPS	30 (20.8)	2 (5.1)	4.87	1.38, 30.98	0.04
CPS	79 (54.9)	18 (46.2)	1.42	0.70, 2.91	0.34
PSG	31 (21.5)	8 (20.5)	1.06	0.46, 2.69	0.89
Absence	19 (13.2)	11 (28.2)	0.39	0.17, 0.92	0.03
Tonic	49 (34.0)	1 (2.6)	19.60	4.05, 352.84	< 0.01
Clonic	3 (2.1)	0 (0.0)		1.00*	
TC	45 (31.3)	10 (25.6)	1.32	0.61, 3.06	0.50
Myoclonic	47 (32.6)	4 (10.3)	4.24	1.58, 14.80	< 0.01
Atonic	13 (9.0)	1 (2.6)	3.77	0.72, 69.57	0.21
Specific epileptic syndrome (%)					
Infantile spasm	24 (16.7)	2 (5.1)	3.70	1.03, 23.69	0.09
Lennox–Gastaut syndrome	17 (11.8)	0 (0.0)		0.03*	
Seizure frequency (%)					
<1/month ^b	8 (5.6)	19 (48.7)	1.00		
2–4/months	28 (19.4)	5 (12.8)	13.30	4.03, 51.58	< 0.001
1–6/week	35 (24.3)	5 (12.8)	16.62	5.11, 63.91	< 0.001
Daily	73 (50.7)	10 (25.6)	17.34	6.28, 52.91	< 0.001
Status epilepticus (%)	34 (23.6)	1 (2.6)	11.75	2.40, 212.18	0.02
Febrile seizures (%)	18 (12.5)	4 (10.3)	1.25	0.43, 4.54	0.70
Family history of seizure (%)	25 (17.4)	8 (20.5)	0.81	0.35, 2.09	0.65
Symptomatic etiology (%)	80 (55.6)	8 (20.5)	4.84	2.17, 11.98	< 0.001
Brain MRI (abnormal) (%)	79 (54.9)	9 (23.1)	4.05	1.86, 9.62	< 0.001
Other clinical manifestations (%)					
MDD	49 (34.0)	3 (7.7)	6.19	2.09, 26.55	< 0.0
Motor deficit	36 (25.0)	2 (5.1)	6.17	1.76, 39.09	0.02
Mental retardation	29 (20.1)	1 (2.6)	9.58	1.94, 173.48	0.03
LDD	29 (20.1)	3 (7.7)	3.03	1.00, 13.15	0.08
Microcephaly	9 (6.3)	1 (2.6)	2.53	0.45, 47.44	0.39
Neonatal Seizure	16 (11.1)	1 (2.6)	4.75	0.92, 87.07	0.14
Hypotonia	6 (4.2)	1 (2.6)	1.65	0.27, 31.69	0.65
Behavioral problem	10 (6.9)	0 (0.0)			0.12*

^a Abbreviations as in Table 3, SPS, Simple Partial Seizure; CPS, Complex Partial Seizure; PSG, Partial Seizure Secondarily Generalized; TC, Tonic-Clonic; MDD, Motor Developmental Delay; LDD, Language Developmental Delay, *P*-values marked by * are obtained by Fisher's exact test because of zero cell.

^b Reference category with which other categories are compared.

Gastaut syndrome were more likely to have intractable epilepsy than the children with benign Rolandic and absence epilepsies.

4. Discussion

This retrospective review of patients with newly diagnosed epilepsy demonstrated that there are a number of

clear factors that are highly associated with medical intractability. Although our findings are preliminary and will need verification with a prospective study, we found strong unilate associations between intractability and several EEG factors: abnormal EEG background including diffuse slowing, asymmetry, abnormal amplitude, a high number of spikes or sharp waves, and focal spike and wave activity. Conversely, reactivity to photic stimulation and 3 Hz spike and wave discharges were predictive of good outcomes in regards to seizure control. With multiple logistic regression, independent predictors of intractability were diffuse slowing and focal spike and wave activity. Strong univariate associates of clinical factors and intractability included simple partial, tonic, and myoclonic seizures; a history of status epilepticus; a symptomatic etiology of the seizures; and an abnormal MRI. Multivariate analysis detected 4 independent clinical features associated with intractable epilepsy:

Table 6
Epileptic syndromes in intractable and well-controlled group

	Intractable %	Controlled %
Infantile spasms (n = 26)	24 (92.3)	2 (7.7)
Lennox–gastaut (n = 17)	17 (100)	0 (0)
Benign rolandic epilepsy (n = 3)	0 (0)	3 (100)
Absence epilepsy (n = 10)	1 (10)	9 (90)

symptomatic etiology; tonic seizures; simple partial seizures; and an early age of onset.

While there have been a number of recent studies designed to evaluate predictors of medical intractability in children, most have concentrated on clinical factors. In 3 large studies dealing with predictors of outcome (Berg et al., 1996; Camfield et al., 1993; Sillanpää, 1993), EEG features were not considered. Since the EEG is routinely performed during the initial phases of diagnosis, it is somewhat surprising that the usefulness of the test as a prognostic tool has not been closely evaluated.

There have been a limited number of studies in which the early EEG has been compared with outcome. Shafer et al. (1988), in a study of 306 residents of Rochester, Minnesota, who developed epilepsy and had an EEG, found that the absence of generalized epileptiform activity was associated with epilepsy remission, defined as 5 years without a seizure. No other EEG features were found to be predictive of outcome. However, in this study, subcategories of generalized spike-and-wave were collapsed, and included atypical spike-wave discharges as well as generalized 3 Hz spike and wave. The authors found that focal epileptiform activity on the first EEG was associated with a good prognosis, perhaps reflecting a high percentage of patients with benign Rolandic epilepsy.

The clinical features that were associated with poor outcomes were similar to the findings of others and were not particularly surprising. It has been noted that the early onset of seizures is an unfavorable prognostic sign (Huttenlocher and Hapke, 1990; Berg et al., 1996). Berg et al. (1996) also associated remote symptomatic etiologies, infantile spasm, and a history of status epilepticus with intractable childhood epilepsy. Sillanpää (1993) examined 40 cases with refractory epilepsy, defined as annual seizures for at least the last 10 years with no more than one year of seizure freedom, and 138 patients with better seizure control. Poor short-term outcome of AED therapy, occurrence of status epilepticus, a high initial seizure frequency, and remote symptomatic seizures were predictors of seizure intractability. In reviewing favorable clinical characteristics predictive of remission, Camfield et al. (1993) reported that seizure onset before 12 years, normal intelligence, lack of neonatal seizures, and fewer than 21 seizures before treatment were predictors of seizure remission. However, they did not include patients with absence seizure, infantile spasm, myoclonic, and atonic seizures.

Our goal in this study was to determine those clinical and EEG findings that are predictive of medical intractability. It is clear that identification of an epileptic syndrome can provide enormous prognostic information. In this study infantile spasms and Lennox–Gastaut syndrome were associated with poor outcomes while absence seizures and benign Rolandic epilepsy was associated with favorable outcomes. However, when a clinician first sees a child with the onset of epilepsy it may be difficult to determine the electroclinical syndrome. This study provides clinical

and electroencephalographic information that may provide the clinician with some guidance.

Some comments regarding the potential weaknesses of this study are necessary. We used a strict criterion for well-controlled epilepsy; that is, being seizure free for one year. It could be argued that children with occasional seizures are well-controlled. However, the goal of most treating physicians is to eliminate seizures totally. There is also a lack of consensus regarding the definition of medical intractability, and our decision to define this group as failing 3 or more AEDs was arbitrary. However, children who fail 3 AEDs have only a 10% chance of achieving a remission of more than one year with subsequent treatments (Carpay et al., 1998).

The conclusions of this study are also limited by the methodology used. The study was retrospective and data was not collected in a systematic fashion. Our results should be confirmed with a prospective study. Nevertheless, the findings demonstrate the value of both the EEG and clinical features in identifying those children destined to develop intractable epilepsy.

Because our study was a retrospective study and Children's Hospital is a tertiary referral hospital, our patient population may be biased toward including patients with severe epilepsy. Nevertheless, the EEG and clinical features we identified as related to medical intractability may be helpful in developing treatment strategies in the future.

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